Course Name: BTech (CSE)

Course Outcome

CO1: Understanding of complexity representation in terms of Big Oh, Theta and Omega notations.
CO2: Derive and solve recurrences describing the performance of divide-and-conquer algorithms.
CO3: Compare and analyze different advance data structures such as RB Tree, B Tree, Binomial Heaps, Fibonacci Heaps.
CO4: Explain the major graph algorithms and their analyses such as DFS, BFS, Dijkstra's Bellman Ford. Employ graphs to model various problems, when appropriate and their analysis.
CO5: Understand the greedy paradigm and able to analyze when an algorithmic design situation calls for it. Synthesize greedy algorithms (Optimal Reliability Allocation, Minimum Spanning Trees, factorial Knapsack) and analyze them.
CO6: Synthesize dynamic-programming algorithms (0/1 knapsack problem, Resource allocation problem, Warshal's and Floyd's algorithms) and analyze them.
CO7: Understand the backtracking and branch & bound paradigm and able to analysis when an algorithmic design situation calls for it.
CO8: Understand the Complexity Theory with P, NP, NP-complete, NP-Hard.

Printed Pages: **University Roll No.** .....

Mid Term Examination, Odd Semester 2023-24
B.Tech. (CSE), III Year, V Semester
BCSC1012: Design and Analysis of Algorithms

Time: 2 Hours **Maximum Marks: 30**

Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL

1	What is an algorithm and why analysis of algorithm is important?
	What is the complexity of the following code?
	for (i=n; i>=1; i=i/2)
	{
	Statement;
	}
		3	CO1	U	F

2	How many comparisons will be made to sort the array A={1,5,3,8,2} using Counting sort? Write an algorithm of Counting Sort.	3	CO2	U	P
3	Solve the following recurrence equations: c) $T(n) = 16\,T(n/4) + n^3$ using Master Method d) $T(n) = \begin{cases} 1 & n = 1 \\ 2T(n/2) + n & n > 1 \end{cases}$ using Recursion Tree	3	CO2	A	P
4	Create a B-tree of Degree 3 for the following list of elements L=86, 50, 40, 3, 94, 10, 70, 90, 110, 113, 116.	3	CO3	U	C

5	Banks often record transactions on an account in order of the times of the transactions, but many people like to receive their bank statements with checks listed in order by check number. People usually write checks in order by check number, and merchants usually cash them with reasonable dispatch The problem of converting time-of-transaction ordering to check-number ordering is therefore the problem of sorting almost sorted input. Argue that the algorithm Insertion-sort would tend to beat the procedure Quicksort on this problem. What is the complexity of the QuickSort algorithm in this situation? Write the Recurrence Relation and solve it.	3	CO2	A	C

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	What are the properties of a red-black tree? Write the steps for the insertion operation in a red-black tree. Construct a red-black tree by inserting the keys in the following sequence into an initialiy empty red-black tree: 13, 10, 8, 3, 4 and 9.	5	CO3	U	C
7	Write the properties of a Binomial Tree. Construct a Binomial Heap for the given list of elements: 7, 2, 4, 17, 1, 11, 6, 8, 15, 10, 20. How a Binomial Heap is different from a Binary Heap.	5	CO3	U	C
8	A CPU scheduling algorithm takes a higher priority job first. Which data structure is used to implement this scheduling algorithm? Write an algorithm for sorting the jobs for this scheduling. Find out the complexity the sorting algorithm.	5	CO2	An	C